Task 7: Session 7 (AIML) - Assignment Questions

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1. **Creating Tuples**

   Write a program to create tuples using different methods:

   - A tuple with 5 numbers using parentheses.
   - A mixed tuple containing integer, string, float, and boolean.
   - An empty tuple using two different ways.
   - A single-element tuple with the value 99.

   Print all tuples and their types. Add comments explaining the rules for single-element tuples.

   Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
(10, 20, 30, 40, 50)
<class 'tuple'>
(42, 'hello', 3.14, True)
<class 'tuple'>
()
<class 'tuple'>
()
<class 'tuple'>
(99,)
<class 'tuple'>
PS D:\Mountreach_ITR_2026>
```

2. Tuple Packing

Create a tuple without using parentheses (tuple packing) to store: your name, age, and city.

Print the tuple and its type.

Then unpack the tuple into three separate variables and print them individually.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
('Mihir', 18, 'Pune')
<class 'tuple'>
Name: Mihir
Age: 18
City: Pune
PS D:\Mountreach_ITR_2026>
```

3. Indexing & Negative Indexing

Create the tuple: colors = ('red', 'green', 'blue', 'yellow', 'purple', 'orange')

Print:

- First element
- Third element
- Last element (using negative index)
- Second last element (using negative index)

Take an index number as input from the user and print the element at that index.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
First Element: red
Third Element: blue
Last Element: orange
Second Last Element: purple
Enter An Index:3
Element At Index: 3 : yellow
PS D:\Mountreach_ITR_2026>
```

4. Slicing Tuples

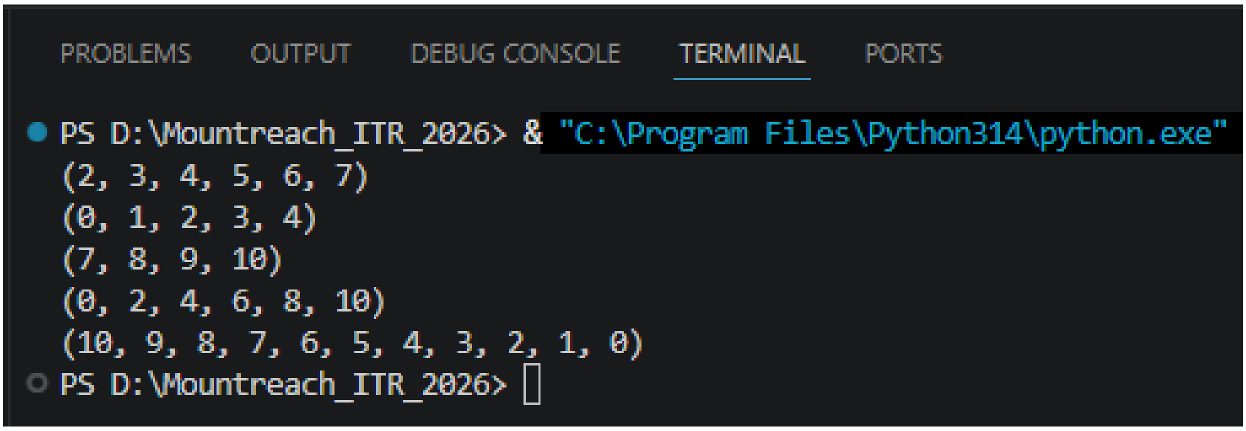
Given the tuple: numbers = (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

Using slicing, print:

- Elements from index 2 to 7
- First 5 elements
- Last 4 elements
- Every second element
- The entire tuple in reverse order

Add comments explaining the slicing syntax.

Output:



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
(2, 3, 4, 5, 6, 7)
(0, 1, 2, 3, 4)
(7, 8, 9, 10)
(0, 2, 4, 6, 8, 10)
(10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0)
PS D:\Mountreach_ITR_2026>
```

5. Nested Tuples

Create a nested tuple to represent a 3•3 matrix:

matrix = ((1, 2, 3), (4, 5, 6), (7, 8, 9))

Print:

- The first row
- The element at second row, third column
- The element at third row, first column

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
First Row: (1, 2, 3)
Second Row, Third Column: 6
Third Row, First Column: 7
PS D:\Mountreach_ITR_2026>
```

6. Iterating & Unpacking

Create a tuple: student = ('Rahul', 20, 'Computer Science', 'A')

- Iterate through the tuple using a for loop and print each item.
- Unpack the tuple into four variables (name, age, branch, grade) and print them with descriptive messages.

Output:

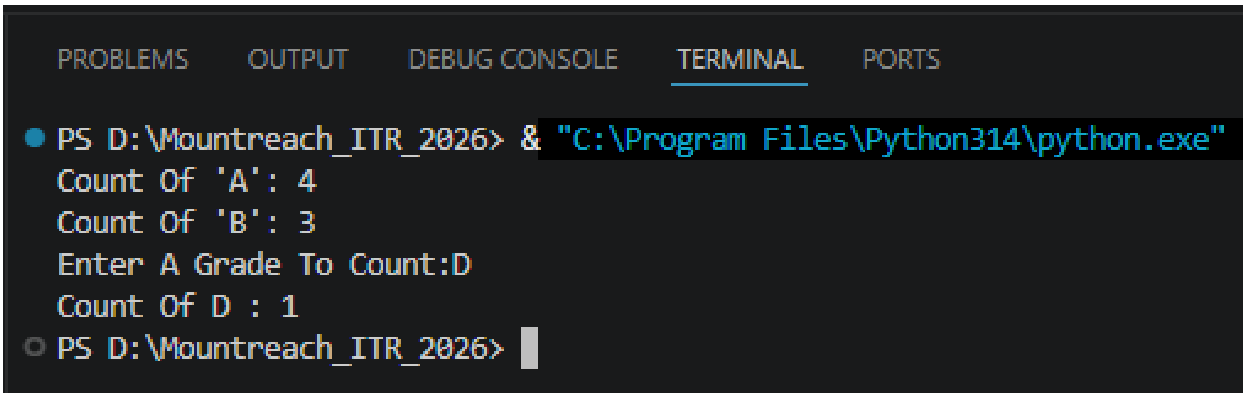
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Rahul
20
Computer Science
A
Name: Rahul
Age: 20
Branch: Computer Science
Grade: A
PS D:\Mountreach_ITR_2026>
```

7. **count() Method**

Create the tuple: grades = ('A', 'B', 'A', 'C', 'A', 'B', 'D', 'A', 'B')

- Count and print how many times 'A' appears.
- Count and print how many times ' B' appears.
- Take a grade as input from the user and print how many times it appears.

Output:



8. index() Method

Create the tuple: fruits = ('apple', 'banana', 'cherry', 'banana', 'mango', 'apple')

- Find and print the first index of ' banana'.
- Find and print the first index of ' banana' starting the search from index 2.
- Use try-except to safely find the index of ' kiwi' and print "Not found" if it doesn' t exist.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
First Index Of 'banana': 1
Index Of 'banana' From Index 2: 3
Not Found
PS D:\Mountreach_ITR_2026>
```

9. Immutability Of Tuples

Create a tuple: coordinates = (10, 20)

Demonstrate that tuples are immutable by trying to:

- Change the first element
- Add a new element using append()

Explain what errors occur (in comments). Then show the correct way to modify data by converting the tuple to a list, making changes, and converting back to a tuple.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe" d:/Mountreach_ITR_2026/Tasks/Task_7/
Traceback (most recent call last):
  File "d:\Mountreach_ITR_2026\Tasks\Task_7\Python_Files\Immutability_Of_Tuples.py", line 3, in <module>
    coordinates[0] = 99
    ~~~~~~~~~~~^^^
TypeError: 'tuple' object does not support item assignment
PS D:\Mountreach_ITR_2026>
```

10. Mini Project - Combined Concepts

Create a **Student Record Program** using tuples:

- Take input for name, roll number, 3 subject marks.
- Store all information in a tuple (use packing).
- Print the complete record.
- Unpack the tuple and print each value with proper labels.
- Use count() to check how many times a particular mark appears (if any).
- Create a nested tuple to store multiple student records (at least 2 students) and access specific values.

Add meaningful comments and use proper formatting.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter Student Name: Mihir Patil
Enter Roll Number: 38
Enter Marks For Subject 1: 97
Enter Marks For Subject 2: 78
Enter Marks For Subject 3: 39

Complete Record: ('Mihir Patil', 38, 97, 78, 39)

Name: Mihir Patil
Roll No.: 38
Subject 1: 97
Subject 2: 78
Subject 3: 39

Enter A Mark To Check Frequency: 89
Mark 89 Appears 0 Time.

All Student Records
('Mihir', 'R01', 85, 90, 78)
('Kunal', 'R02', 92, 88, 95)

Second Student's Name: Kunal
First Student's Subject 2 Marks: 90
PS D:\Mountreach_ITR_2026>
```